

Fluent but Fragile: Failures to Maintain Shared Task Interpretation in Task-Based Dialogue with Large Language Models

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Abstract

Large language models (LLMs) are increasingly used in task-based dialogue, such as planning, scheduling, and analytical decision support, where users often infer successful collaboration from fluent and cooperative responses. In this paper, we present a focused empirical study showing that such fluency can mask failures to maintain shared task interpretation across turns. We examine two task-based dialogue settings—constraint-based scheduling and definition-driven risk analysis—across multiple LLMs. While some models consistently satisfy constraints in scheduling dialogues, they do not make their interpretation of those constraints visible to the user. In contrast, definition-driven dialogues exhibit systematic breakdowns across all models, with failures manifesting as polite clarification deferral rather than hallucination or refusal. These findings highlight an interactional visibility problem and motivate the design of task-based dialogue systems that make assumptions, constraints, and interpretations explicit to support more reliable human-AI collaboration.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Natural language interfaces*; • **Computing methodologies** → *Artificial intelligence*.

Keywords

task-based dialogue, large language models, human-AI collaboration, shared task interpretation, multi-turn interaction

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1 Introduction

Large language models (LLMs) are increasingly used in *task-based dialogue*, where users interact with a system over multiple turns to accomplish concrete goals such as planning, scheduling, or analytical decision support. In these settings, users often rely on conversational cues—fluency, politeness, and responsiveness—to judge whether the system is collaborating effectively. Prior work in human-computer interaction has shown that such surface cues can strongly shape user trust and reliance on automated systems,

sometimes independently of actual system understanding or correctness [3, 7].

Task-based dialogue, however, requires more than locally appropriate responses. Effective collaboration depends on maintaining a shared interpretation of the task across turns, including its goals, constraints, and definitions. The importance of making understanding visible has long been emphasized in theories of grounding and common ground, which argue that collaborators must continuously provide evidence of mutual understanding in order to coordinate successfully [2]. In interactive systems, insufficient feedback about system reasoning can impair users' ability to form accurate mental models and assess alignment during interaction [1].

Consider a user engaging an LLM to assess risks associated with deploying an academic advising chatbot. In an initial turn, the user provides an explicit definition of “key risk,” specifying that each risk must include a compliance obligation, a plausible failure mode, affected stakeholders, and a mitigation control. The model acknowledges the definition and appears to understand the task. In a subsequent turn, the user asks for a concise executive summary. Rather than applying the previously provided definition, the model responds by requesting additional clarification, reframing the task as underspecified. The response is fluent and cooperative, yet the original task is not completed. From the user's perspective, it is unclear whether the system failed to understand the definition, chose not to apply it, or is intentionally deferring. Importantly, no overt error occurs, and the interaction remains conversationally appropriate.

Most prior work on failures in language-based systems emphasizes outcomes that are readily observable, such as hallucinated content or incorrect answers [4]. While these failures are important, they do not capture all of the ways interaction can break down in task-based dialogue. Problems may instead emerge gradually, with the system producing locally appropriate responses while failing to preserve task-defining elements across turns.

Recent work on human-AI collaboration highlights the risks of overreliance on fluent AI systems, particularly when users lack visibility into system reasoning or assumptions [1, 6]. In conversational settings, this lack of visibility can make it difficult for users to determine whether apparent task success reflects shared understanding or merely locally plausible behavior.

In this paper, we examine the gap between apparent task success and shared understanding in task-based dialogue with LLMs. We present a focused empirical study of two common task types—constraint-based scheduling and definition-driven risk analysis—that require persistent maintenance of task interpretation across

turns. Our results show that fluent dialogue does not reliably signal shared task interpretation, motivating the design of task-based dialogue systems that better communicate their assumptions, constraints, and interpretations to users.

2 Related Work

Our work draws on prior research in grounding, trust in automation, and failure modes of large language models, with a focus on how understanding is made visible—or remains hidden—during interaction.

Grounding theory emphasizes that effective collaboration depends on participants providing evidence of shared understanding over time [2]. In task-oriented dialogue, such evidence allows collaborators to coordinate goals, constraints, and meanings, and to detect and repair misunderstandings when they arise. Prior work in human-computer interaction has shown that when systems fail to provide adequate feedback about their reasoning or internal state, users may struggle to form accurate mental models and assess whether mutual understanding has been achieved [5]. These challenges become more pronounced in multi-turn interaction, where task interpretation must be preserved across successive contributions.

Research on trust and reliance in automation further shows that users often rely on surface cues such as fluency, confidence, and responsiveness when judging system competence [3, 7]. More recent work in human-AI interaction highlights the importance of transparency and explanation in helping users calibrate trust appropriately, particularly when systems are capable of producing persuasive but potentially misleading outputs [1, 6]. In conversational settings, a lack of visibility into system assumptions or interpretations can encourage overreliance, even when underlying task alignment has degraded.

Failures of large language models have most commonly been studied in terms of hallucination or factual incorrectness [4]. While these failure modes are important, they do not fully capture the interactional breakdowns observed in task-based dialogue. In our study, models often avoid producing incorrect content by deferring or reframing the task, resulting in failures of shared task interpretation without explicit error or refusal. This highlights the need to study interactional dynamics beyond output correctness, particularly in settings where users must assess alignment over extended dialogue.

Our work contributes to this line of research by providing empirical evidence that fluent and cooperative dialogue can mask failures to maintain shared task interpretation across turns, underscoring the importance of making task understanding visible in human-AI collaboration.

3 Method

We conducted a controlled, multi-turn interaction study to examine how shared task interpretation is maintained—or fails to be maintained—over the course of task-based dialogue with LLMs. Our goal was not to benchmark performance, but to surface recurring interactional patterns that emerge across dialogue turns. Each trial

consisted of a short, fixed-length conversation in which a user attempted to complete a task through successive prompts to a single model.

3.1 Task Design Rationale

We selected two task types that are common in real-world task-based dialogue and that require different forms of persistent interpretation across turns.

Constraint-based scheduling tasks required the system to construct a schedule subject to explicit temporal constraints, such as non-overlapping events and mandatory breaks, while accommodating evolving user preferences. In our study, these tasks were framed as organizing a short conference program. Successful completion depends on maintaining all stated constraints across turns, rather than solving a single-shot optimization problem.

Definition-driven risk analysis tasks required the system to apply a formally specified definition consistently across turns. In these tasks, the user provided an explicit definition of “key risk” that included multiple required elements: a compliance obligation, a plausible failure mode, affected stakeholders, and a mitigation control. Subsequent turns requested reframing or summarization while implicitly requiring that the original definition remain operative.

These tasks were chosen because failures in task interpretation can occur even when individual responses appear reasonable or helpful, making breakdowns difficult to detect during interaction.

3.2 Dialogue Structure

Each interaction consisted of three user turns following an initial system instruction. Later turns were designed to introduce opportunities for divergence in task interpretation without explicitly contradicting earlier instructions. For example, in the risk analysis task, users requested executive summaries or reformulations that implicitly tested whether the original definition was preserved.

This structure mirrors realistic task-based dialogue, where users often refine or adapt requests rather than restating constraints or definitions verbatim.

3.3 Models and Execution

We evaluated three widely used LLMs from two providers. Each model was tested on both task types across eight independent trials using identical prompts, resulting in 24 trials per task. All interactions were conducted in a deterministic or low-temperature setting when supported, to reduce variability unrelated to task interpretation.

We did not provide corrective feedback during interaction. This allowed us to observe whether models spontaneously maintained or externalized task interpretation across turns.

3.4 Operationalizing Failures of Shared Task Interpretation

We operationalized failures as cases where task-defining elements were not preserved across dialogue turns despite fluent and cooperative responses.

For scheduling tasks, failures included violations of explicit constraints (e.g., missing breaks or overlaps) as well as cases where constraints were satisfied without being explicitly acknowledged or

User: Define the key risks of deploying an academic advising chatbot. A key risk must include (1) a compliance obligation, (2) a plausible failure mode, (3) affected stakeholders, and (4) a mitigation control.

Assistant: Understood. I will identify risks that meet all four criteria.

User: Please summarize the top three risks for an executive audience.

Assistant: To do this properly, I need clarification on which regulations apply (e.g., FERPA, ADA, GDPR) and the jurisdiction of deployment. Once confirmed, I can proceed.

Figure 1: Example from a definition-driven task-based dialogue. Although the assistant acknowledges the task definition, it later reframes the task as underspecified rather than applying the provided definition.

confirmed. For definition-driven tasks, failures included responses that did not apply the user-provided definition, including cases where the system deferred task completion by reframing the task as underspecified.

We do not treat stylistic variation or alternative valid formulations as failures. Our analysis is conservative and centers on preservation of the user’s stated task framing across turns.

4 Results

We report results separately for the two task types, followed by a cross-task comparison. Figure 1 illustrates a representative interaction from the definition-driven task, highlighting how failures to maintain shared task interpretation can remain interactionally hidden. Table 1 summarizes observed failure rates across models and tasks.

4.1 Definition-Driven Risk Analysis

Failures to maintain shared task interpretation were uniform in the definition-driven risk analysis task. Across all three models and all eight trials per model ($n = 24$), none of the interactions completed the task as defined by the user. In every trial, the model failed to apply the user-provided definition of “key risk” across turns, despite having acknowledged that definition earlier in the dialogue.

Instead of generating risks that satisfied the specified criteria—a compliance obligation, a plausible failure mode, affected stakeholders, and a mitigation control—models consistently reframed later requests as underspecified. Follow-up prompts requesting executive summaries or reformulations triggered clarification requests for additional information such as applicable jurisdiction, specific regulations, or whether internal institutional policies qualified as compliance obligations.

The excerpt in Figure 1 illustrates this pattern. After restating the definition accurately, the model introduces new prerequisites and conditions task completion on further user input, rather than

Table 1: Failure rates in maintaining shared task interpretation across tasks and models.

Task	Provider	Model	Failure Rate	n
Risk definition	Gemini	gemini-2.5-flash	1.0	8
Risk definition	OpenAI	gpt-4.1	1.0	8
Risk definition	OpenAI	gpt-5.2	1.0	8
Scheduling	Gemini	gemini-2.5-flash	0.5	8
Scheduling	OpenAI	gpt-4.1	0.0	8
Scheduling	OpenAI	gpt-5.2	0.0	8

Table 2: Observed interactional failure patterns across tasks.

Failure Pattern	Scheduling (n=24)	Risk Def. (n=24)
Clarification deferral	0	24
Constraint violation	4	0
Constraint-satisfying (no grounding)	20	0

proceeding with the analysis as defined. This response is fluent, polite, and framed as cooperative, yet it does not execute the task. Importantly, this breakdown does not take the form of hallucination or explicit refusal. Instead, the dialogue remains locally coherent while task execution is deferred, leaving no clear signal to the user that the task definition has been abandoned or altered.

4.2 Constraint-Based Scheduling

Constraint-based scheduling produced more heterogeneous outcomes. One model violated explicit constraints in half of the trials (4/8), typically by omitting required breaks or introducing overlaps. The remaining two models produced schedules that satisfied all explicit constraints across all trials (0/8 failures each).

However, even in cases where constraints were satisfied, models did not explicitly restate or confirm the constraints they were enforcing. As a result, scheduling interactions exhibited a second failure pattern: opaque success. Although the final output was valid, users received no interactional evidence that constraints were being tracked intentionally rather than coincidentally. This makes it difficult for users to assess whether shared task interpretation has been achieved.

4.3 Task Structure Effects

Comparing task types reveals a strong effect of task structure on failures to maintain shared task interpretation. In definition-driven tasks, failures were systematic across all models and trials (24/24). In contrast, constraint-based scheduling exhibited variable outcomes, including both explicit constraint violations and superficially successful outputs that lacked interactional grounding. These results suggest that failures to maintain shared task interpretation are driven more by task structure than by model choice.

Failure patterns were identified through manual inspection of interaction transcripts and grouped by interactional outcome. Aggregating across all scheduling trials ($n = 24$), explicit constraint violations occurred in 4/24 runs (16.7%). The remaining 20/24 runs

produced constraint-satisfying schedules but did not provide interactional evidence that constraints were being tracked intentionally, reinforcing the distinction between output correctness and observable shared task interpretation.

5 Discussion

Our results show that fluent, cooperative dialogue is not a reliable signal of shared task interpretation in task-based interaction with LLMs. Across both task types, models produced responses that appeared reasonable at the level of individual turns while failing to preserve task-defining elements across the dialogue.

In definition-driven tasks, clarification requests functioned as a form of interactional deferral. Although clarification is often treated as desirable behavior, in this context it consistently prevented task completion despite the presence of an explicit definition. These responses remained conversationally appropriate and non-hallucinatory, making the breakdown difficult to detect from surface dialogue alone.

In constraint-based tasks, failures took a different form. Even when models satisfied all explicit constraints, they did not externalize their interpretation of those constraints. As a result, successful outcomes were interactionally opaque: users received no evidence that constraints were being tracked intentionally rather than coincidentally. This opacity becomes more consequential as tasks grow longer or more complex, where undetected misalignment can accumulate over time.

Together, these findings suggest that task-based dialogue systems should not rely on offering fluent responses alone, but should make task interpretation visible throughout interaction. Design directions include explicitly restating active constraints or definitions, distinguishing clarification from deferral, and supporting repair of task interpretation through confirmation prompts or inspectable task representations.

We emphasize that this work is intentionally exploratory. Rather than proposing a comprehensive taxonomy of failures, our goal is to highlight a recurring interactional pattern that is difficult to detect yet consequential in practice, and to motivate further investigation into how task interpretation can be made observable and repairable in human-AI dialogue.

6 Limitations and Future Work

This study focuses on system behavior and interactional outcomes rather than user experience. We did not evaluate whether users notice these failures during interaction, how quickly they detect them, or how such failures influence trust and reliance over time. Prior work suggests that users often rely on surface cues such as fluency when assessing AI competence, raising the possibility that these interactional failures may go unnoticed until downstream consequences emerge.

Future work should examine how users interpret clarification deferral and opaque success in task-based dialogue, and whether these behaviors lead to overtrust or delayed error detection. Controlled user studies could measure users' ability to detect breakdowns in shared task interpretation and assess how different interface cues—such as explicit restatement of constraints or definitions—affect trust calibration and repair behavior.

More broadly, this work points to the need for evaluation frameworks that consider not only output correctness, but also whether users can reliably assess and respond to task interpretation failures during interaction.

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